TOBIAS LADNER

Curriculum Vitae

Current Position

04/2026 – present
Postdoctoral Researcher
Technical University of Munich, Germany
AI Interpretability and Automated Reasoning

Education

02/2022 – 03/2026
Ph.D. • **Formal AI Safety**
Technical University of Munich, Germany
with *"summa cum laude"*

09/2025 – 02/2026
Visiting Researcher
University of California, Irvine, USA

10/2019 – 12/2021
Master of Science
Technical University of Munich, Germany
Computer Science • with high distinction

Winter 2020
Erasmus Exchange Semester
University of Sussex, United Kingdom
Computer Science

10/2016 – 08/2019
Bachelor of Science
Technical University of Munich, Germany
Computer Science • with merit

Selected Publications

[1] **On Convex and Non-Convex Abstraction Refinement Techniques Guaranteeing Safety of Artificial Intelligence**. Tobias Ladner. *Ph.D. Thesis. Technical Unviersity of Munich*. 2026.

[2] **The Misery of Mechanistic Interpretability: A Formal Perspective**. Tobias Ladner and Matthias Althoff. *arxiv*. 2026.

[3] **Provably Explaining Neural Additive Models**. Shahaf Bassan*, Yizhak Yisrael Elboher*, Tobias Ladner*, et al.. *ICLR*. 2026.

[4] **Out of the Shadows: Exploring a Latent Space for Neural Network Verification**. Lukas Koller, Tobias Ladner, and Matthias Althoff. *ICLR*. 2026.

[5] **Explaining, Fast and Slow: Abstraction and Refinement of Provable Explanations**. Shahaf Bassan*, Yizhak Yisrael Elboher*, Tobias Ladner*, Matthias Althoff, and Guy Katz. *ICML*. 2025.

[6] **Fully Automatic Neural Network Reduction for Formal Verification**. Tobias Ladner and Matthias Althoff. *TMLR*. 2025.

Professional Contributions and Activities

- Administrator of toolbox CORA
- Repeatability co-chair of ARCH-COMP (2023-2026) and VNN-COMP (2025-2026)
- Invited talks at Munich Datageeks, University of Aalborg, and at LEISYS: "Formal Verification of Neural Networks in Safety-Critical Environments"
- Hobby projects: Android Reddit Wallpaper, Covid Data Visualization

About Me
Tobias Ladner
Munich, Germany

Specialization
AI Safety,
Formal Methods,
Adversarial Robustness,
Automated Reasoning,
Interpretability,
AI Alignment

Programming
python, matlab, lean,
java, kotlin, ocaml,
latex, html, css, js;
agentic programming

Languages
German (native),
English (C2)

toladner@pm.me
Website
LinkedIn
Google Scholar
GitHub